

Coordination-Driven Terpyridine-Based Twisted Prisms with Tunable Emissions and Hierarchical Self-Assembly Properties

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Tunable fluorescence emission materials play a critical role in many fields such as biology and chemistry. With respect to metallo-supramolecular assemblies, the construction of 3D tunable-fluorescent supramolecular assemblies with 2,2':6',2''-terpyridine (tpy) is still in its infancy. In this study, two luminous 3D supramolecular prisms self-assembled with two terpyridine-based tetraphenylethylene (TPE) ligands and metal Zn(II) are prepared, and the influence of hydrophobic alkyl chains on tunable fluorescence is further explored. By exploiting hydrophobic properties, the prism SA exhibits tunable visible-light emission in the molecular-aggregated state. Moreover, SA forms left-hand and right-hand helical nanofibers by hierarchical self-assembly. However, the prism SB without hydrophobic chains does not exhibit tunable fluorescence emission, and only random spherical aggregates after hierarchical self-assembly are formed. This study can aid to further strengthen the understanding about the effect of hydrophobic segments on photophysical properties and hierarchical self-assembly of discrete supramolecular architectures.

1. Introduction

In the past decades, tunable-fluorescence materials have attracted extensive attention and exhibited a wide range of applications in biological imaging,^[1] environmental sensing,^[2] fluorescent probes,^[3] and other fields,^[4] due to their high sensitivity and stimulus responsiveness.^[5] What is more,

tunable-fluorescence materials, which are affected by external factors, such as temperature,^[6] force,^[7] light,^[8] and solvent,^[9] and internal factors, such as donor–receptor,^[10] π – π interaction,^[11] and host–guest interaction,^[12] play a key role for the development of light-emitting materials. However, the studies on fluorescent-tunable materials by hydrophobic interactions are scarce yet. In addition, conventional organic fluorophores are well known to typically exhibit weakening or even completely quenching (aggregation-caused quenching, ACQ) in a concentrated solution or in the solid state, which limited several practical applications of these fluorophores.^[13] To solve undesirable ACQ, Tang and co-workers have developed a series of fluorescent species exhibiting the opposite effect known as aggregation-induced emission (AIE).^[14] As one of the superior iconic AIE fluorophores, tetraphenylethylene (TPE) and its derivatives have been widely synthesized and investigated, due to their simple molecular structure and special geometry.^[15]

Coordination-driven self-assembly has been proven to be effective for constructing supramolecular coordination complexes (SCCs),^[16] because metal coordination bonds not only provide definite directionality and high predictability but also have moderate bond energies (15–25 kcal mol^{−1}) to ensure self-modulation and thermodynamic stability during self-assembly.^[17] If chromophores are combined with SCCs that have the specific stoichiometry and precise geometry, these complexes can be suitable for exploring luminescent properties. Most of the currently reported luminescent SCCs have been constructed using pyridyl-based ligands with different styles of chromophores.^[9c,17b,18] Among coordination-driven self-assembly, 2,2':6',2''-terpyridine (tpy),^[19] which is a donor with a remarkable binding ability to diverse transition-metal ions with different binding strengths,^[20] has been widely used as a versatile building block to construct various discrete supramolecular architectures, such as 2D metallomacrocycles,^[21] 3D polyhedra,^[22] and prisms.^[23] However, only a few SCCs have been constructed by the combination of terpyridine and fluorescent groups.^[24] Moreover, suitable linkers combining the fixed 180° tpy-M(II)-tpy connectivity with functional fluorescent groups hinder to the design of emissive 3D supramolecular

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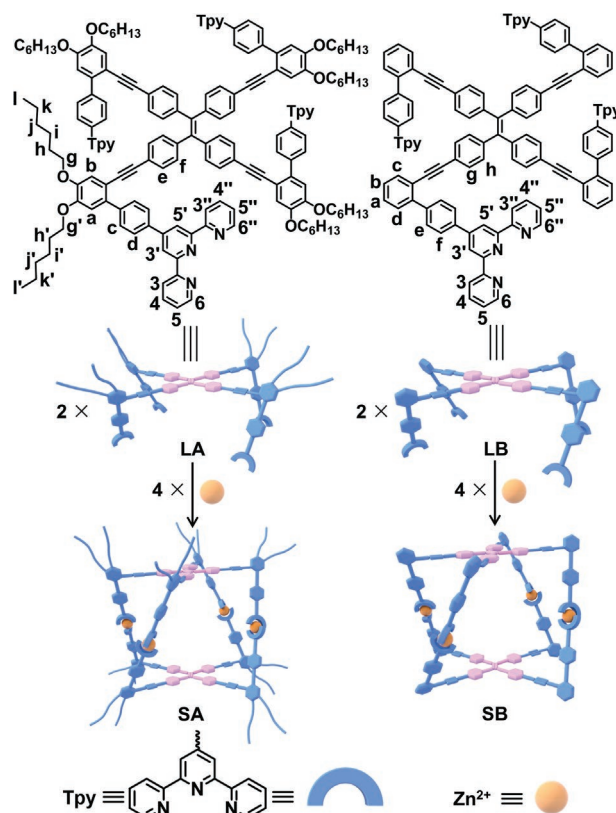
architectures. Therefore, the construction of 3D fluorescent-tunable supramolecules based on tpy building blocks still remains challenging.

Herein, we reported two twisted prisms SA and SB assembled two terpyridine-based TPE ligands LA (with eight hydrophobic C₆ alkyl chains) and LB (without alkyl chains) with Zn(II), respectively. Ligands LA and LB were with an angle of 60° between the TPE core and four terpyridine units. And the photophysical properties of two twisted prisms were investigated in dilute solutions and aggregated states. Owing to the introduction of multiple hydrophobic chains in the vertexes of ligand LA, prism SA exhibited significantly tunable visible-light emission in the molecular aggregated states and in the different polarity of solvents. On the contrary, prism SB without alkyl chains only exhibited a slight change in the aggregated state and in the presence of different polar solvents. Moreover, hierarchical self-assembly behavior was further examined. From TEM images, prism SA underwent hierarchical self-assembly to form left-hand and right-hand helical nanofibers. Different from the hierarchical self-assembly of SA, spherical aggregates with diameters of 9–30 nm were observed in the SB system. All of the aforementioned phenomena clarified that the introduction of multiple hydrophobic chains could render tunable emissions and hierarchical self-assembly properties.

2. Results and Discussion

As shown in Scheme S1, Supporting Information, ligands LA and LB with four terpyridine units and one TPE core were synthesized by Sonogashira coupling reactions in a satisfactory yield (66% for LA and 80% for LB) and purified by column chromatography (SiO₂). Compared with LB, ligand LA had eight hydrophobic C₆ alkyl chains introduced on its vertexes. Structures of LA and LB were completely characterized by nuclear magnetic resonance (NMR) spectroscopy, including ¹H (Figures S1,S5, Supporting Information), ¹³C (Figures S2,S6, Supporting Information), 2D correlation spectroscopy (2D-COSY) (Figures S3,S4,S7, Supporting Information), and matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrometry (Figures S19,S20, Supporting Information). Prisms SA and SB were successfully prepared by the self-assembly of two ligands and Zn(NO₃)₂·6H₂O with a precise stoichiometric ratio of 1:2 in a mixed solvent of CHCl₃ and MeOH (1:3, v/v) at 50 °C for 10 h, followed by the addition of excess NH₄PF₆ to afford yellow precipitates in a high yield (90% for SA and 92% for SB) (Scheme 1).

The formation of prisms SA and SB was initially characterized by NMR spectroscopy (Figure 1A,B; Figures S8–S11,S13–S15, Supporting Information). After coordination, broadened peaks were observed in the ¹H NMR spectra of SA and SB, suggesting complexes were obtained. The tpy-*H*^{3',5'} protons of the terpyridine were significantly shifted downfield due to the decrease in the electron density after coordination. On the contrary, all protons of tpy-*H*^{6,6''} were expected to upfield due to the electron-shielding effect (Figure 1A,B).^[19c] In addition, according to 2D-COSY results (Figures S10,S11,S15, Supporting Information), only one set



Scheme 1. Self-assembly of prisms SA and SB.

of proton signals of tpy was displayed, demonstrating the formation of highly symmetrical structures without other isomers. Furthermore, 2D diffusion-ordered spectroscopy (DOSY) in CH₃CN revealed a single band at log *D* = −9.39 for SA and −9.40 for SB, supporting the inference of the formation of a single discrete structure (Figures S12,S16, Supporting Information). The diameters of these two prisms derived from the log *D* values were 3.04 and 2.98 nm, which were in agreement with the simulated values of SA and SB calculated by the Stokes–Einstein equation,^[25] respectively.

Electrospray ionization mass spectrometry (ESI-MS) and traveling-wave ion mobility mass spectrometry (TWIM-MS) have been employed to validate the clean formation of expected assemblies. Only one set of continuous peaks with different charge states in SA and SB systems were observed from ESI-MS spectra (Figure 1C,D), due to the successive loss of different numbers of PF₆[−] counterions. After deconvolution, the molecular weight of these two prisms were 6938.3 Da (SA) and 5336.9 Da (SB), which were consistent with the expected chemical compositions of two ligands and four metal Zn(II). Each isotope pattern of these peaks matched well with the corresponding simulated isotope pattern (Figures S21,S22, Supporting Information). For further characterization, TWIM-MS was employed to provide additional evidence for the proposed structures SA and SB. Single species of charged signals with narrow drift-time distributions were exclusively observed (Figure 1E,F), indicating the formation of discrete supramolecules SA and SB

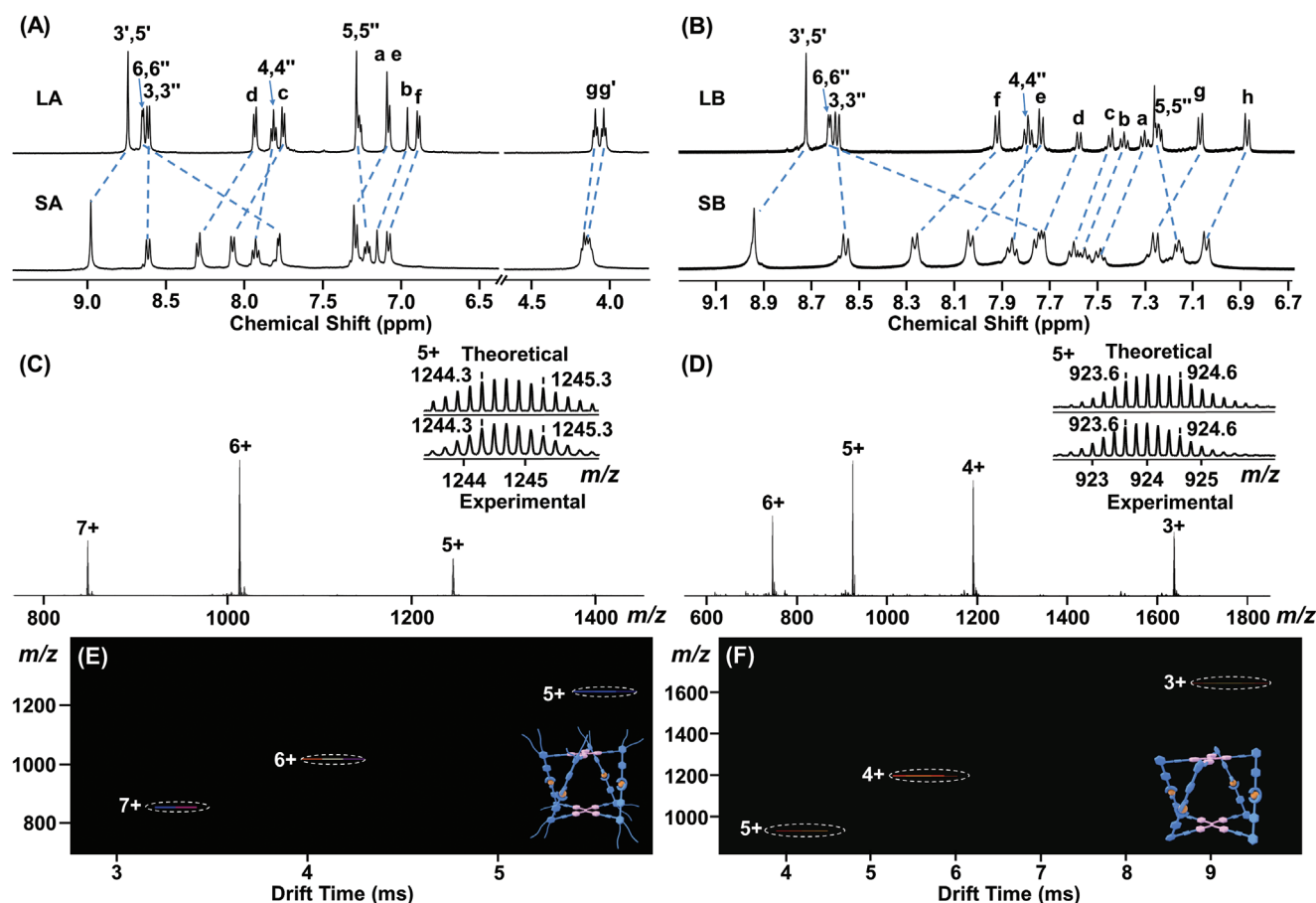


Figure 1. ^1H NMR spectra (500 MHz, 300 K) of A) LA in CDCl_3 and prism SA in CD_3CN and of B) LB in CDCl_3 and prism SB in CD_3CN . ESI-MS of C) SA and D) SB and 2D TWIM-MS plot (m/z vs drift time) of E) SA and F) SB.

without any superimposed fragments or overlapping isomers or conformers.

After the successful characterization of the structural composition of these ligands and complexes, their absorption and luminescent properties were examined in chloroform and acetonitrile (Figure 2A,B). Ligands LA and LB in chloroform ($c = 1.00 \mu\text{M}$) exhibited the main absorption peaks at 260 and 280 nm, originating mainly from tpy and TPE and a long-wavelength absorption band centered at 385 nm attributed mainly to intramolecular charge transfer. Compared with the ligands, the main absorption peaks of SA and SB were bathochromically shifted by ≈ 20 and 50 nm (280 and 320 nm), respectively, due to the metal to ligand charge transfer (MLCT) after coordination. These assemblies exhibited a significantly enhanced molar absorption coefficient due to the inclusion of two ligands in one metallacage structure.^[6f] For ligands LA and LB, the emission peak at 430 nm in the fluorescence spectrum, which should be ascribed to the tpy group. Prism SA exhibited an emission band at 550 nm, corresponding to the TPE group. However, owing to the incomplete charge transfer from the metal to the ligand, in the fluorescence spectrum of SA, the emission of the tpy group at 425 nm was reduced but not completely quenched. Unlike that observed for SA, the emission peak of SB was observed at ≈ 510 nm, while the emission

of tpy group was completely quenched by MLCT after coordination with Zn(II) . Because the absorption and emission of the charge-transfer process from metal to ligand (MLCT) of SA were destroyed, the intensity and absolute quantum efficiency of its fluorescence emission were less than those of SB. The Φ_{em} value of SB in acetonitrile was 7.89%, which was significantly greater than the Φ_{em} of SA in acetonitrile (2.89%) and greater than that of ligand LB in chloroform (5.7%). Therefore, subtle changes were made by the reduction of the alkyl chain in the ligand, which could significantly increase the emission of complexes in dilute solutions.

Next, the optical properties of prisms SA and SB were evaluated in different solvents. First, 2D diffusion-ordered spectra in different solvents were employed to confirm prisms SA and SB were indeed in monodisperse state (Figures S17,S18, Supporting Information). Photophysics of the two prisms in dilute solution revealed an irregular change on the increase in the solvent polarity (Figure 3A,B). Prism SA spanned three distinct colors depending on solvents (Figure 3A), suggesting that the emission was affected by intramolecular charge transfer caused by the hydrophobic chain segment. In a low-polarity solvent such as THF, a red color was observed. However, an orange color was observed in moderate-polarity solvents, such as acetone and CH_3CN . In high-polarity solvents such as

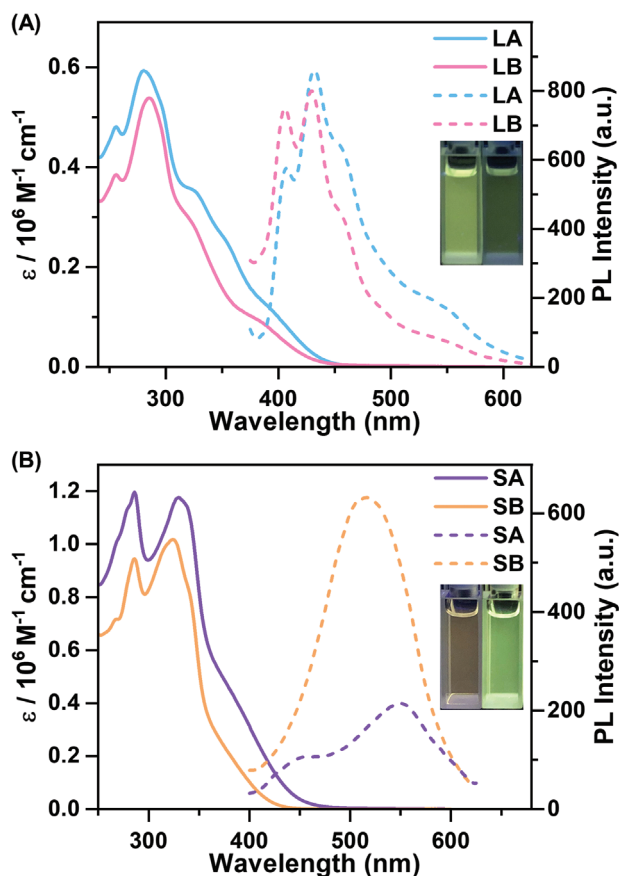


Figure 2. A) UV-vis and PL spectra of LA and LB in CHCl_3 ($\lambda_{\text{ex}} = 320 \text{ nm}$, $c = 1.00 \mu\text{M}$). B) UV-vis and PL spectra of SA and SB in acetonitrile ($\lambda_{\text{ex}} = 320 \text{ nm}$, $c = 1.00 \mu\text{M}$).

DMSO and DMF, a light yellow color was observed. As prism SB did not dissolve in THF, only fluorescence spectra of SB in the other four solutions were investigated. Because SB did not comprise alkyl chains, the change in solvent polarity would hardly affect intramolecular charge transfer, the emission wavelength changed slightly, and the overall color was cyan (Figure 3B).

To further understand the relationship between emission behavior and hydrophobic motifs, fluorescence spectra were recorded in a mixed solvent of acetonitrile and water (Figure 4A,B). The introduction of water into an acetonitrile solution reduced the solubility of the components, thereby promoting aggregate formation. For prism SA in Figure 4A, with the constant increase in the water content, the tpy emission peak disappeared, and the intensity of the TPE emission peak gradually increased. At a water content of 30%, the fluorescence intensity reached the maximum, which was consistent with the AIE behavior. Owing to the interaction of the alkyl chains between molecules, the electron-donating ability was reduced, resulting in a certain blue-shift of the spectrum. With a water content of greater than 50%, the formation of aggregates led to the extremely tight packing of the supramolecules, resulting in ACQ. In addition, after coordination, the transfer of charge from the metal to the ligand would result in the red-shift of

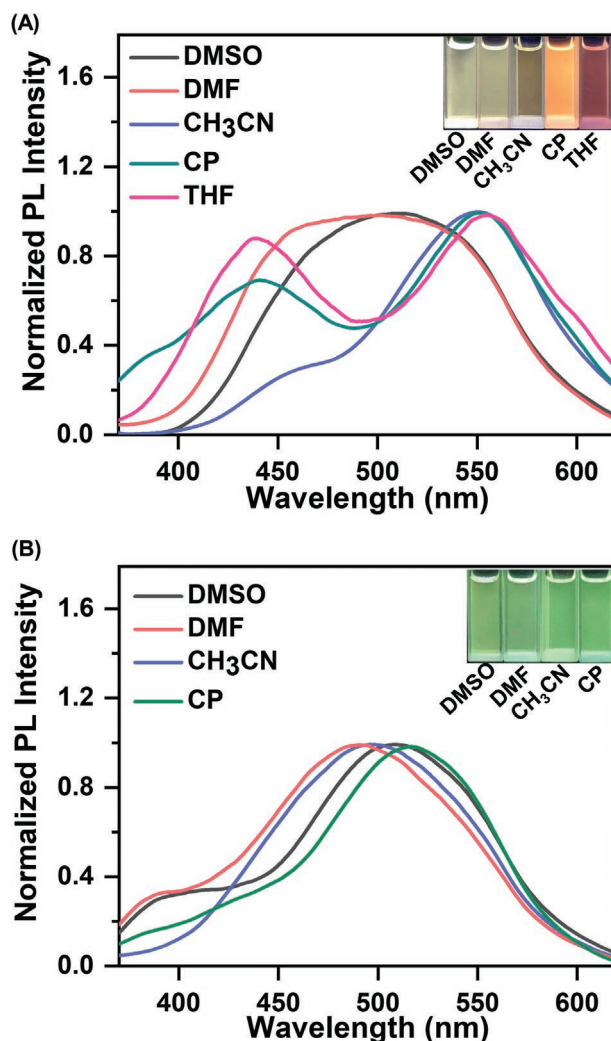


Figure 3. Solvent effect on the light-emitting properties of SA and SB ($\lambda_{\text{ex}} = 320 \text{ nm}$, $c = 1.00 \mu\text{M}$). DMSO, dimethyl sulfoxide; DMF, *N,N*-dimethylformamide; CH_3CN , acetonitrile; CP, acetone; THF, tetrahydrofuran.

the emission of the AIE fluorophore in the polar solution and attenuation in less polar environment. When water content of greater than 50%, owing to the increasing polarity of solution, the emission wavelength displayed red-shift. The solution fluorescence changed from orange fluorescence to yellow-green fluorescence; finally, red fluorescence was observed by excitation at 365 nm with an ultraviolet lamp at 298 K (inset in Figure 4A). Similar to SA, SB also exhibited a certain AIE behavior, followed by ACQ behavior. Throughout the titration, the Φ_F value agreed well with the spectrum, showing an initial increase and then decrease (Figure S25, Supporting Information). The difference was that the fluorescence intensity of SB reached its maximum at a water content of 50%. In the aggregated state, the increased fluorescence intensity of SB was observed to be 5.8 times, while that of SA was only 4.3 times. This should be attributed to the hydrophobic alkyl chain of SA resulted in the worse solubility in water than that of SB, which promoted to form large aggregates of SA further lead to the decrease in the luminous

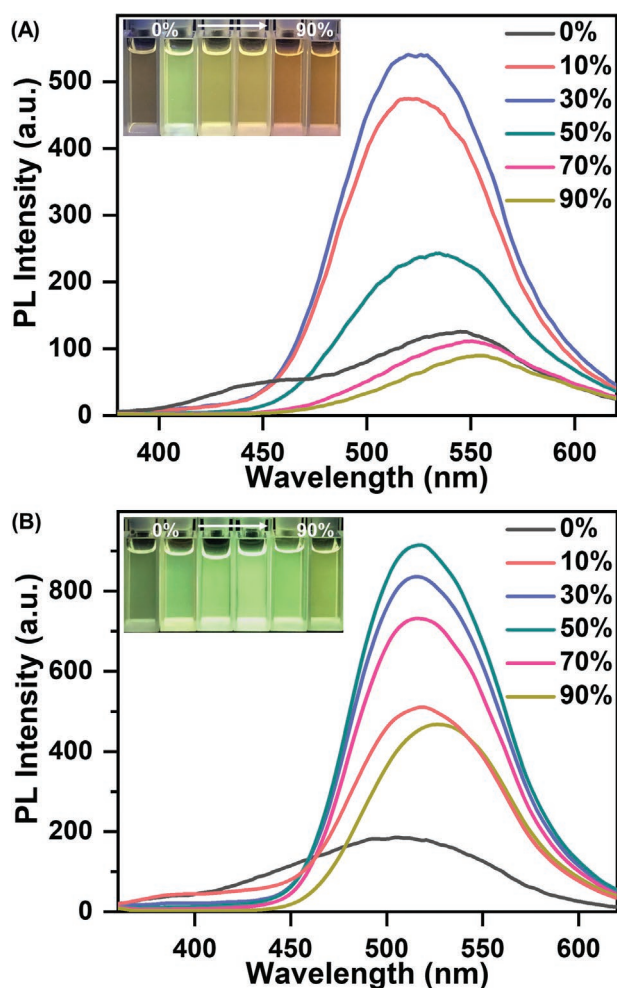


Figure 4. PL spectra of A) SA and B) SB versus H₂O in CH₃CN/H₂O mixtures (λ_{ex} = 320 nm, c = 1.00 μM).

intensity.^[26] At the same time, hydrophobic segments were helpful for the formation of regular aggregates, also causing a higher red-shift in the emission wavelength. However, for SB, it was speculated to form random aggregates, and it exhibited a relatively weak shift in its fluorescence emission wavelength; hence, the solution always exhibited yellow-green fluorescence (inset in Figure 4B).

Inspired by different spectral phenomena of the two prisms after the introduction of poor solvents, these two prisms were speculated to also form aggregates with different morphologies after hierarchical self-assembly. Hence, the dynamic study on the hierarchical self-assembly behavior of prisms SA and SB was studied via transmission electron microscopy (TEM). All samples were obtained by the diffusion of isopropyl ether into the solution of the prisms (1 mg mL⁻¹ in DMF) and deposition of the solution on the surface of ultrathin carbon lacey support films on a copper grid. From TEM images of SA, a regular hierarchical self-assembly behavior was not clearly observed by the diffusion of isopropyl ether for 2 days and 4 days. However, on the 6th day, scattered helical nanofibers started to appear with diffusion of the poor solvent

(Figure S24, Supporting Information). On the 8th day, a large number of entangled nanofibers were observed (Figure 5A). From the images with enlarged scales, nanofibers showed a certain helicity (left-handed shown in Figure 5B, right-handed shown in Figure 5C). These fibrous structures had widths of 21 nm and pitches of 27 nm. The hierarchical self-assembly result of prism SA was possibly attributed to the following two aspects: i) the helical primitives from single supramolecule SA; ii) hydrophobic interactions from intermolecular alkyl chains.^[27] The twisted prisms were stacked with a certain direction offset through the interaction between the hydrophobic chains to form a single helix. The helices were further gathered to form a helicoid with a width of about 21 nm (Figure 5E). Different from the hierarchical self-assembly behavior of SA, spherical aggregates with diameters of 9–30 nm were observed in the SB system (Figure 5D). These observations suggested divergent aggregation behavior between SA and SB would be affected with the existence of alkyl chains. In addition, this result further indicated that the hydrophobic segments could help to form regular linear aggregates.

3. Conclusion

In summary, two prisms SA and SB were successfully assembled with two tetrapotic 2,2':6',2''-terpyridine-based TPE ligands and Zn(II) by changing hydrophobic motifs, and the two prisms with completely different luminescence properties were clearly exploited in different solvents and aggregation states. Compared with prism SB without the alkyl chains, prism SA with 16 hydrophobic C6 alkyl chains exhibited tunable fluorescence emission in different solvents and aggregated states. This would help to strengthen further understanding about the effect of hydrophobic segments on photophysical properties. In addition, inspired with the completely different photophysical properties in the aggregated state, the hierarchical self-assembly of two prisms was further investigated. From the TEM images, SA formed spiral nanofibers with hydrophobic interactions of alkyl chains, while SB formed spherical aggregates. Clearly, the hydrophobic segment could aid in the formation of an orderly stack. This study also provided a new opportunity to regulate the layered self-assembly behavior of self-assembled metal complexes.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

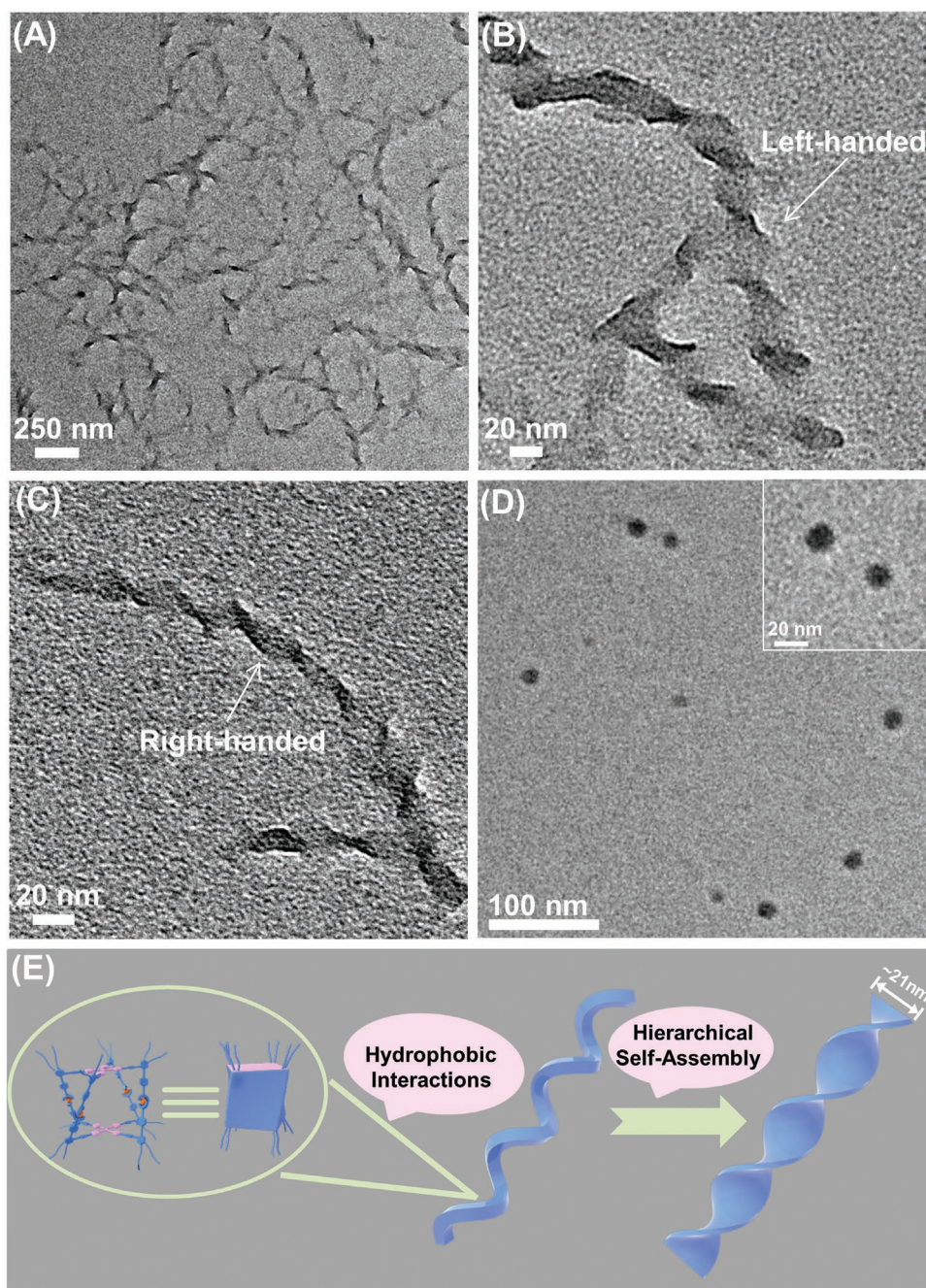


Figure 5. TEM images of hierarchical self-assembly formed by A–C) SA and D) SB. E) Proposed mechanism for hierarchical self-assembly of SA.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

hierarchical self-assembly, supramolecular self-assembly, terpyridine, tetraphenylethylene

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